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Course : DSA0204 - Computer Vision

CO1 - AT1

1) How does Sampling affect Image Quality

Sampling is the process of converting a Continuous Image into pixels The Quality of the image depends on the Sampling rate If Sampling rate is low, the image becomes blurred and looses fine details

→ Determines the number of pixels in an image
→ low Sampling causes blur
→ High Sampling improves clarity

2) Image Sensing and acquisition in low light

Image Sensing and acquisition capture light from Scene using Camera Sensors In low-light Conditions less light reaches the Sensor, resulting in Dark and noisy images

→ Sensor recieves less light
→ Image appears dark
→ Fine Details are lost
→ Noice increases
→ Better Lightning improves image Quality

3) Cause of aliasing artifacts

Aliasing occurs when an image is sampled below the required rate. It produces jagged edges and Distorted patterns Increasing the Sampling rate and using anti - aliasing filters can reduce this problem

→ Caused by low Sampling
→ Creates Jagged edges
→ Distorts image details
→ use anti - aliasing filter

4) Pixel resolution vs Intensity Resolution

Pixel resolution represents the number of pixels in an image while intensity resolution represents the number of brightness levels Both are important for producing clear and Detailed image

→ pixel resolution affects Sharpness
→ More pixels give better detail
→ intensity resolution Controls brightness
→ Higher intensity improves Contrast

5) Level of Computer vision in Survellance

A Survellance System uses high-level Computer vision because it not only Captures images but also recognize Objects and makes Decisions automatically

→ uses High-level Computer vision

→ Detects moving Objects

→ Recognize people

→ Supports real time monitoring

6) Image formation in Autonomous Driving

Image formation explains how light from a Scene reaches the Camera Sensor proper image formation improves image clarity, helping autonomous vehicles accurately detect lanes, Signs and obstacles

→ Improves image clarity

→ Reduces Distortion

→ Detects lanes Correctly

→ Identifies Obstacles

→ Increases Driving Saftey

7) Role Of Quantization

Quantization Converts Continious Intensity values into Discreate Digital Value. Higher Quantization level represents more image details. While lower levels reduce image Quality

→ Converts analog to Digital values
→ Produce digital images
→ preserves brightness information
→ lower levels causes Detail loss

8) Noise During Image Aquisition

Noise is introduced during image acquisition due to poor lightning, Sensor limitations or electronic infeence It can be reduced using better Sensors proper lightning

→ Caused by low light
→ poor Sensors increase noise
→ use Quality Sensors

9) Levels Of Computer Vision in facial recognition

Facial recognisition uses Different levels of computer vision. It first Captures the image then extracts facial features and finally Compares them with Sorted Data to identify

→ low level : Image aquisition
→ Mid level : Feature extraction
→ High level : Face recognisition

10) Sampling and image formation

Object visibility Depends on proper Sampling and image formation. low Sampling reduces the image resolution, while poor image formation Causes blur making objects difficult to Detect